

Project Charter

Project Title

AI-Powered Credit Risk Analytics & Loan Default Prediction Platform

Project Overview

Financial institutions must balance business growth with effective risk management. Poor lending decisions can result in loan defaults, financial losses, and reduced profitability. Home Credit serves customers with limited or non-traditional credit histories, making credit risk assessment more challenging.

This project aims to develop a data-driven Credit Risk Analytics Platform that leverages analytics, machine learning, business intelligence, and AI to predict loan defaults, identify high-risk borrowers, and support smarter lending decisions.

Business Problem

Home Credit has access to extensive customer, credit bureau, and repayment data, but transforming this information into actionable risk insights remains a challenge.

Business Question:

How can historical customer and credit data be used to predict loan default risk and improve lending decisions?

Without an effective risk assessment framework, the organization may experience:

- Higher default rates
- Increased financial losses
- Poor portfolio performance
- Inefficient lending decisions

Project Objectives

Primary Objectives

- Predict loan default probability.
- Identify high-risk borrowers.
- Improve credit approval decisions.
- Reduce credit-related losses.
- Enhance portfolio risk visibility.

Secondary Objectives

- Understand key drivers of default.
- Develop customer risk segmentation.
- Build a risk scoring framework.
- Generate AI-powered risk insights.
- Demonstrate end-to-end analytics capabilities.

Business Objectives

The project supports the following goals:

- Reduce credit risk exposure.
- Improve portfolio quality.
- Increase operational efficiency.
- Enable data-driven decision-making.
- Improve profitability through better lending practices.

KPI Framework

Portfolio KPIs

- Total Loan Applications
- Total Approved Loans
- Default Rate
- Portfolio Exposure

Customer KPIs

- Average Customer Income
- Average Loan Amount
- Credit History Length
- High-Risk Customer Count

Risk KPIs

- Risk Score
- Delinquency Rate
- Previous Default Rate

Model KPIs

- Accuracy
- Precision
- Recall
- F1 Score

- ROC-AUC

Analytical Approach

Descriptive Analytics

Analyze customer, credit, and loan behavior.

Diagnostic Analytics

Identify factors contributing to defaults.

Predictive Analytics

Build machine learning models to estimate default probability.

Prescriptive Analytics

Provide recommendations for lending decisions and risk mitigation.

AI-Assisted Analytics

Generate explainable risk assessments and customer-specific recommendations.

Key Deliverables

Business Deliverables

- Credit Risk Assessment Framework
- Customer Risk Segmentation
- Business Recommendations

Technical Deliverables

- SQL Analysis
- Exploratory Data Analysis
- Statistical Analysis
- Machine Learning Models
- Power BI Dashboard
- AI Risk Assistant
- Streamlit Application

Documentation Deliverables

- Business Understanding Documents
- Domain Knowledge Repository
- Final Executive Report
- GitHub Project Repository

Success Criteria

The project will be considered successful if it:

- Identifies major drivers of loan default.
- Accurately predicts high-risk customers.
- Provides actionable business recommendations.
- Delivers interactive Power BI dashboards.
- Demonstrates advanced SQL, Python, Machine Learning, and AI skills.

High-Level Roadmap

1. Business Understanding
2. Domain Knowledge
3. Data Understanding
4. Data Preparation
5. SQL Analytics
6. Exploratory Data Analysis
7. Feature Engineering
8. Statistical Analysis
9. Machine Learning
10. Explainable AI
11. Power BI Dashboard
12. AI Risk Assistant
13. Business Recommendations
14. Final Reporting

Expected Business Value

The platform will help stakeholders:

- Improve lending decision accuracy.
- Reduce loan default losses.
- Enhance portfolio quality.
- Strengthen risk management capabilities.
- Increase operational efficiency.
- Support transparent and data-driven decision-making.

By combining analytics, machine learning, business intelligence, and AI, the project provides a scalable framework for modern credit risk management and responsible lending practices.